

MAHAN BALOOEI

Bologna, Italy · mahanbalooei2@gmail.com · LinkedIn · GitHub · Website

RESEARCH INTERESTS

- Protein language models & foundation models
- Machine learning for biological data
- Variant-effect & pathogenicity prediction
- Computational genomics & multi-omics

EDUCATION

University of Bologna — MSc in Bioinformatics

Oct 2024 – Present

MSc Thesis: *Protein Language Models for Human Protein Variant Pathogenicity Prediction*

Supervisors: Castrense Savojardo, Pier Luigi Martelli, Matteo Manfredi

Ongoing thesis on modernizing the E-SNPs&GO framework for missense variant pathogenicity prediction through updated human variant datasets and contemporary protein language models.

Iran University of Medical Sciences — BSc in Medical Laboratory Sciences

Sept 2019 – Jul 2023

Admitted through Iran's National University Entrance Examination, placing in approximately the **top 1%** among 750,000 applicants.

Relevant Coursework: Molecular Diagnostics, Immunoassays, Clinical Chemistry, Haematology, Microbiology, Virology.

PUBLICATIONS

Farmani, S.*, Dehghanbanadaki, N.*, **Balooei, M.***, Mehralitabar, H., Sabaghian, H., & Naderi-Manesh, H. **Mechanism-based targeting of lactate dehydrogenase tetramerization by naturally derived compounds: A survey of molecular docking, molecular dynamics, and QSAR analysis.** *Computers in Biology and Medicine*, 197(Pt B), 111000, 2025.

*These authors contributed equally.

- [Article link](#)

RESEARCH EXPERIENCE

Protein Language Model-Based Signal Peptide Prediction

Apr 2026

MSc Bioinformatics Project – Independently Implemented

- Explored whether pretrained protein representations could improve eukaryotic signal-peptide prediction compared with classical sequence-based and task-specific neural approaches.
- Built a high-confidence UniProtKB/Swiss-Prot dataset and used MMseqs2 clustering to reduce redundancy and limit homology-driven leakage between training and benchmark sets.
- Compared von Heijne PSWM, biochemical-feature SVM, CNN-BiLSTM, and ESM-2 models; the ESM-2-based approach achieved the strongest held-out performance ($F1 = 0.962$, $MCC = 0.957$).
- Examined amino-acid composition, kingdom-specific differences, and cleavage-site motifs to check whether the model's performance was consistent with known biological patterns.
- [GitHub Repository](#)

Protein Domain and Functional Annotation Using HMMER and MMseqs2

Feb 2026 – Mar 2026

- Compared profile-based and sequence-similarity approaches for protein-domain classification to understand where each method is most effective.
- Built HMMER- and MMseqs2-based workflows using structure-derived alignments and evaluated both approaches under the same classification setting.

- Both methods performed strongly (MCC = 0.997 and 0.991), highlighting broader questions around remote homology and protein-family classification.
- [GitHub Repository](#)

Computational Epigenomics Analysis of DNA Methylation (Illumina 450K)

Jun 2025 – Jul 2025

- Analyzed differential DNA methylation patterns while using quality control, normalization, and PCA to distinguish biological variation from technical effects.
- Identified differentially methylated regions and interpreted the resulting patterns using heatmaps, volcano plots, and exploratory statistical analysis.
- [GitHub Repository](#)

Chest X-Ray Pneumonia Detection via Transfer Learning

Apr 2026 – May 2026

- Compared general-purpose and radiograph-specific pretrained models to assess how transfer-learning choices affect pneumonia classification under class imbalance.
- Tested threshold selection, ensembling, test-time augmentation, and Grad-CAM; DenseNet-121 XRV gave the strongest performance under the maximum-MCC strategy.
- [GitHub Repository](#)

SKILLS

Protein Bioinformatics: ESM-2, protein representation learning, HMMER, MMseqs2, BLAST, UniProt, PDB, protein functional/domain annotation, sequence-function prediction.

Machine Learning & Deep Learning: PyTorch, scikit-learn, TensorFlow, SVMs, CNN-BiLSTM, transfer learning, representation learning, model evaluation, class-imbalance handling.

Functional Genomics & Epigenomics: R/Bioconductor, minfi, Illumina 450K methylation analysis, differential methylation, PCA-based batch-effect assessment.

Research Computing: Python, R, Bash/Linux, SQL, Git/GitHub, Docker, HPC/Slurm, conda, Jupyter Notebooks.

ADDITIONAL EXPERIENCE

Clinical Laboratory Internships

Tehran, 2022–2023

Rotations in hematology, microbiology, and clinical chemistry across hospital and diagnostic laboratory settings.

LANGUAGES

English: Professional proficiency

Persian (Farsi): Native

Italian: Basic communication